

Standard Form & Indices

Test 7 · 41 marks total · Sections F & G include Higher-tier content

Name: _____

Date: _____

Score: _____ / 41

Section A — Laws of indices

A1. Write $9^3 \times 9^6$ as a single power of 9.

[1 mark]

A2. Write $4^{13} \div 4^8$ as a single power of 4.

[1 mark]

A3. Write $(8^2)^5$ as a single power of 8.

[1 mark]

A4. Work out the value of $10^2 \times 10^1$.

[2 marks]

Section B — Zero and negative indices

B1. Work out the value of 58^0 .

[1 mark]

B2. Work out the value of 9^{-2} . Give your answer as a fraction.

[2 marks]

B3. Work out the value of 2^{-2} . Give your answer as a decimal.

[2 marks]

Section C — Writing standard form

C1. Write 29,000 in standard form.

[2 marks]

C2. Write 4,500,000 in standard form.

[2 marks]

C3. Write 0.0081 in standard form.

[2 marks]

C4. Write 0.0000006 in standard form.

[2 marks]

Section D — Converting back, and calculating

D1. Write 1.8×10^6 as an ordinary number.

[1 mark]

D2. Write 5×10^{-3} as an ordinary number.

[1 mark]

D3. Work out $(9 \times 10^2) \times (4 \times 10^6)$. Give your answer in standard form.

[3 marks]

D4. Work out $(4 \times 10^8) \div (8 \times 10^2)$. Give your answer in standard form.

[3 marks]

Section E — Powers to know, and estimating roots

E1. Write down the value of 5^4 .

[1 mark]

E2. $\sqrt{40}$ lies between which two consecutive whole numbers?

[1 mark]

Section F — Laws of indices with algebra

F1. Simplify $d^7 \times d^8$.

[1 mark]

F2. Simplify $e^{16} \div e^{10}$.

[1 mark]

F3. Simplify $(t^4)^4$.

[1 mark]

F4. Simplify $9k^4 \times 2k^3$.

[2 marks]

F5. Simplify $(2w^5)^4$.

[2 marks]

Section G — Fractional indices (Higher)

G1. Work out the value of $144^{1/2}$.

[1 mark]

G2. Work out the value of $343^{1/3}$.

[1 mark]

G3. Work out the value of $9^{3/2}$.

[2 marks]

G4. Work out the value of $1000^{-2/3}$.

[2 marks]